

AMENDMENT NO. 1 JULY 2025
TO
IS 17081 : 2019 AVIATION TURBINE FUEL (KEROSENE TYPE, JET A-1)
CONTAINING SYNTHESIZED HYDROCARBONS — SPECIFICATION

(Page 2, clause 3.2.3) — Substitute the following for the existing:

‘3.2.3 Antioxidants

An approved antioxidant or mixture of antioxidants shall be added to fuel (or fuel component) and/or synthetic blend components which have been severely hydroprocessed to prevent peroxidation and gum formation. The total concentration of active material(s) in the final batch shall not exceed 24.0 mg/l (defence requirements 17 mg/l to 24 mg/l for severely hydro processed component in fuel). For synthetic blend components, requirement for total concentration of active material(s) is 17 mg/l to 24 mg/l.

The point of anti-oxidant addition shall be before or during release from the manufacturing site. Where a finished fuel comprises a blend of synthetic blend components, severely hydro processed, hydro processed, and non-hydro processed components, the requirements for mandatory addition of antioxidant applies only to that portion of the blend which is synthetic blend component and which has been severely hydro processed. In such cases the proportion of the blend which is synthetic blend component and which has been severely hydro processed shall be reported.

The concentration of antioxidant added to the fuel shall be reported as follows:

- a) Report the anti-oxidant concentration as a total active material in the final batch of fuel in refinery certificate of quality. Same shall not exceed 24.0 mg/l;
- b) Report the total antioxidant addition as an equivalent concentration of active material in the proportion of the final batch of fuel that comprises synthetic blend components on the certificate of quality. Same shall 17 mg/l to 24.0 mg/l; and
- c) In case of defense supply, additionally, report the total antioxidant addition as an equivalent concentration of active material in the proportion of the final batch of fuel that comprises severely hydro processed and/or synthetic blend components on the certificate of quality.

NOTE — If the antioxidant is added during ship loading the reporting requirements shall be the same as those in (a), (b) and (c) above.

If anti-oxidant is added to the fuel report the anti-oxidant concentration as a total active material in the final batch of fuel. To account for residual tank heel concentrations of anti-oxidant, where new batches do not contain anti-oxidant, it is not required to report the concentration of active material in the final batch once the calculated dosage is < 0.5 mg/l.

If anti-oxidant is added post manufacture, report the total active material concentration in the final batch of fuel on the certificate of analysis.

NOTE — Antioxidants are mandatory for synthetic blend components and shall be added prior to or during release of the synthetic blend component as per this standard from the designated manufacturing site.

The following are some of the approved antioxidants:

<i>Sl No.</i>	<i>Formulation</i>	<i>Qualification Reference</i>
(1)	(2)	(3)
i)	2,6-ditertiary-butyl-phenol	RDE/A/606
ii)	2,6 ditertiary-butyl-4-methyl-phenol	RDE/A/607
iii)	2,4-dimethyl-6-tertiary-butyl-phenol	RDE/A/608

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Table (Concluded)

<i>Sl No.</i>	<i>Formulation</i>	<i>Qualification Reference</i>
(1)	(2)	(3)
iv)	75 percent minimum, 2,6-ditertiary-butyl-phenol 25 percent maximum, tertiary and tritertiary-butyl-phenols	RDE/A/609
v)	55 percent minimum, 2,4-dimethyl-6-tertiary-butyl-phenol 15 percent minimum, 4 methyl-2,6-ditertiary-butyl-phenol remainder, 30 percent maximum, as a mixture of monomethyl and dimethyl-tertiary-butyl-phenols	RDE/A/610
vi)	72 percent minimum, 2,4-dimethyl-6-tertiary-butyl-phenol 28 percent maximum, mixture of tertiary-butyl-methylphenols and tertiary-butyl dimethyl phenols	RDE/A/611

(Page 3, clause 3.2.6.2, Note) — Substitute the following for the existing:

*NOTE — Some of the qualified lubricity improving additives along with their concentration in which they are added to the fuel at the time of delivery to the purchaser are as follows:

<i>Sl No.</i>	<i>Product</i>	<i>Manufacturer</i>	<i>Qualification Reference</i>	<i>Minimum mg/l</i>	<i>Maximum mg/l</i>
(1)	(2)	(3)	(4)	(5)	(6)
i)	Hitec 580	Afton chemical Ltd	RDE/A/661	15	23
ii)	Octel DCI-4A	Innospec LLC	RDE/A/662	9	23
iii)	Nalco 5403	Nalco chemical Co	RDE/A/664	12	23

[Page 6, Table 1, Sl No. (i)(c) and (d)] — Substitute the following for the existing:

<i>Sl No.</i>	<i>Characteristic</i>	<i>Requirement</i>	<i>Method of Test, Ref to [P:] of IS 1448/ASTM/ IP/ISO</i>
(1)	(2)	(3)	(4)
c)	Particulate contamination, at point of manufacture, mg/l, <i>Max</i> or	1.0 ²⁾	ASTM D 5452*/IP 423
d)	Particulate, at point of manufacture, cumulative channel particle counts:	Individual channel counts	ISO code ³⁾ IP 565*/IP 577/ D7619
	1) $\geq 4 \mu\text{m(c)}$	Report	19, <i>Max</i>
	2) $\geq 6 \mu\text{m(c)}$	Report	17, <i>Max</i>
	3) $\geq 14 \mu\text{m(c)}$	Report	14, <i>Max</i>
	4) $\geq 21 \mu\text{m(c)}$	Report	Report
	5) $\geq 25 \mu\text{m(c)}$	Report	Report
	6) $\geq 30 \mu\text{m(c)}$	Report	13, <i>Max</i>

[Page 6, Table 1, Sl No. (ii)(f)(2)] — Delete.

[Page 7, Table 1, Sl No. (vi)(b)] — Substitute the following for the existing:

<i>Sl No.</i>	<i>Characteristic</i>	<i>Requirement</i>	<i>Method of Test, Ref to [P:] of IS 1448/ASTM/ IP/ISO</i>
(1)	(2)	(3)	(4)
vi)	b) Silver strip corrosion ¹²⁾ classification, for 4 h at 50 °C, <i>Max</i>	1	Annex F

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[Page 8, Note 3, under Table 1] — Substitute the following for the existing:

‘3 Both number of particles and number of particles as a scale number as defined under Table 1 of ISO 4406 shall be reported. If channel counts/ISO code exceed the limits stated, Annex B of IP 565 or Annex B of IP 577 or Annex A2 of ASTM D7619 may be applied to eliminate trace free water, and cleanliness re-determined. In such cases, results before and after application of Annex B of IP 565 or Annex B of IP 577 or Annex A2 of ASTM D7619 shall be reported. To meet the requirements of this standard the limits of either test c or d shall be met.’

[Page 8, Note 4, under Table 1] — Substitute the following for the existing:

‘4 Defence requirements to be met at 22 percent by volume, maximum. Typically, the ASTM D1319 and IP 156 are specified as the required test method for aromatics measurement. In addition, recently delivered supplies of the product gel containing the dye with lot numbers 3000000975, 3000000976, 3000000977, 3000000978, 3000000979 and 3000000980 were produced with a substitute dye that is unfortunately not suitable and will not provide accurate measurements of aromatic concentration if utilized. Given the above, if ASTM D1319/IP 156 is used for analysis of Aromatics, the lot number of the gel used shall be reported on the test certificate. A ruggedness study has shown that subsequent batches of the dye, with a lot number of 3000000983 or greater, yield results comparable with the lots prior to 3000000975. An inter-laboratory study (ILS) is expected to corroborate these results.

When the aromatic level is needed to be determined, Jet A-1 fuel will only meet the aviation fuel operating limitations of aircraft certificated to operate on Jet A-1 fuel and the requirements of IS 1571 if:

- a) the fuel has been tested for aromatics concentration in accordance with IS 1448 (Part 23)/ASTM D1319/IP 156 with a dye other than from lot numbers 3000000975 through 3000000982; and
- b) the fuel has been tested for aromatics concentration in accordance with the alternative test methods ASTM D6379/IP 436.

No other alternative test method or method of deriving the aromatic content, is acceptable.’

[Page 8, Note 8, under Table 1] — Substitute the following for the existing:

‘8 Refinery components used in the make-up of the batch shall be reported on the refinery certificate of quality as a percentage by volume of the total fuel in the batch. Severely hydroprocessed components are defined as those petroleum derived hydrocarbons that have been subjected to a hydrogen partial pressure of greater than 7 000 kPa (70 bar or 1 015 psi) during manufacture.’

[Page 8, Note 13, under Table 1] — Substitute the following for the existing:

‘13 Heater tubes other than those supplied by the original equipment manufacturer (OEM) shall not be used. Technically suitable heater tubes are PAC - Alcor, or Falex. The annexes referred to in Table 1 and this note correspond to those in IP 323. If the technically equivalent ASTM D3241 test method is used, the same protocol shall be followed using the appropriate annex that corresponds to the visual (VTR), interferometric (ITR) or ellipsometric (ETR) method. Tube deposit ratings shall be measured by IP 323 Annex C ITR or Annex D ETR, when available. If the Annex C ITR device reports “N/A” for a tube’s volume measurement, the test shall be a failure and the value reported as > 85 nm. Visual rating of the heater tube shall be by the method in IP 323. Annex B is not required when Annex C ITR or Annex D ETR deposit thickness measurements are reported. In case of dispute between results from visual and metrological methods, the metrological method shall be considered the referee.’

[Page 9, Note 32, under Table 2] — Insert the following at the end:

‘33 During downstream distribution, if the freezing point of the fuel is very low and cannot be determined using IS 1448 (Part 11)/IP 16/ASTM D2386 lowest achievable temperature of - 65 °C, if no crystals appear during cooling of the fuel and when the thermometer indicates a temperature of - 65 °C, the freezing point shall be recorded as below - 65 °C. This limit does not apply if the freezing point is measured by alternate methods mentioned in Annex G.

34 Compliance of requirement for identified Incidental Materials shall be in line with IS 1571 Amendment No. 1.’